RIG-FLAT MAPS OF NOETHERIAN ADIC RINGS ARE STABLE UNDER COMPOSITION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof checks the composite is adic topologically finite type, localizes at each f, and composes the naive rig-flat maps using stability of flatness under composition

Editorial note. The label lemma-composition-rig-flat-continuous names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the arrow property rig-flat on WAdm^Noeth (lemma-rig-flat-axioms of the same chapter) and the composition property of Formal Spaces, Remark formal-spaces-remark-composition-variant-Noetherian
- composition of adic topologically finite type maps stays adic topologically finite type (lemma-composition-finite-type of the same chapter)
- naive rig-flatness of the localizations A_{f} -> B_{f} and B_{f} -> C_{f} follows from rig-flatness
- flat ring maps are stable under composition (Algebra, Lemma algebra-lemma-composition-flat)

Source reference. Original lemma 00HC: sources/stacks-chow/tags.txt, source lines 638–638. Prerequisite pointer invoked at proof line(s) 4073; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 0GBK: sources/stacks-chow/tags.txt, source lines 20375–20375. Prerequisite pointer invoked at proof line(s) 4059; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 01T3: sources/stacks-restricted/pinned-original-source.tex, source lines 2533–2533. Prerequisite pointer invoked at proof line(s) 4067; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GGU: sources/stacks-restricted/pinned-original-source.tex, source lines 4010–4010. Prerequisite pointer invoked at proof line(s) 4063; it names a standard fact of the same chapter that the counted argument

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uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *The property $P(\varphi) =$"φ is rig-flat" on arrows of $WAdm^{Noeth}$ is stable under composition as defined in Formal Spaces, Remark 0GBK.*

Proof. The statement makes sense by Lemma 0GGU. To see that it is true assume we have rig-flat morphisms $A \to B$ and $B \to C$ in $WAdm^{Noeth}$. Then $A \to C$ is adic and topologically of finite type by Lemma 01T3. To finish the proof we have to show that for all $f \in A$ the map $A_{\{f\}} \to C_{\{f\}}$ is naively rig-flat. Since $A_{\{f\}} \to B_{\{f\}}$ and $B_{\{f\}} \to C_{\{f\}}$ are naively rig-flat, it suffices to show that compositions of naively rig-flat maps are naively rig-flat. This is a consequence of Algebra, Lemma 00HC. □